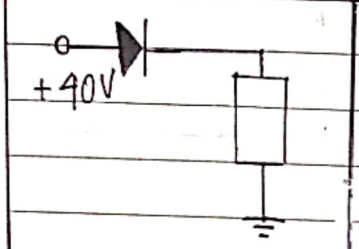
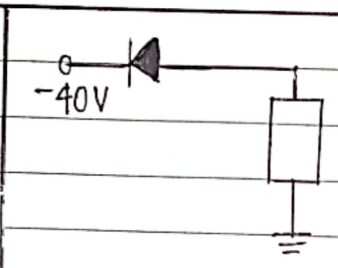
Regita Rasya Zafarana
231344088 1C-TNK

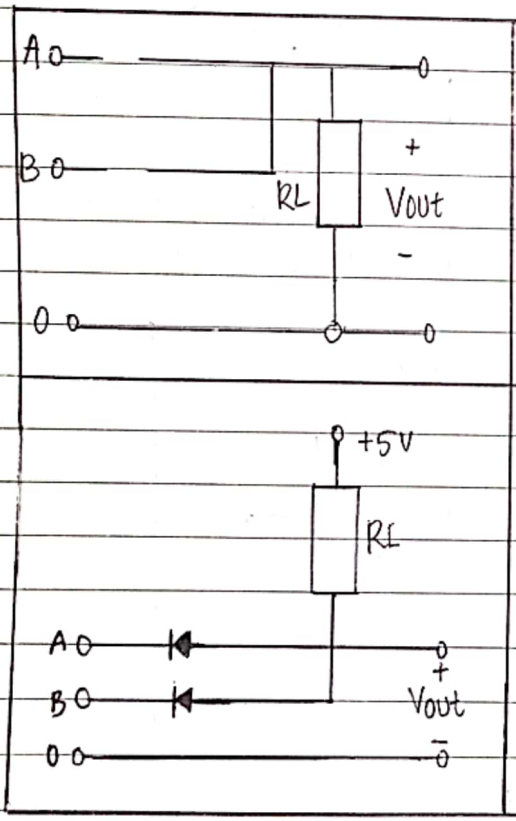
Latihan Soal - Soal Dioda

No
Date

1. Tentukan keadaan masing-masing dioda dengan kondisi bias DC di bawah ini (konduksi atau menyumbat)

			
Kondisi Dioda reverse	Kondisi Dioda forward	kondisi dioda reverse	kondisi dioda forward

2. Asumsikan Dioda Ideal. Tentukan logika keluaran tiap rangkaian dengan berbagai kemungkinan tegangan masukan, kemudian tentukan jenis gerbang logikanya.



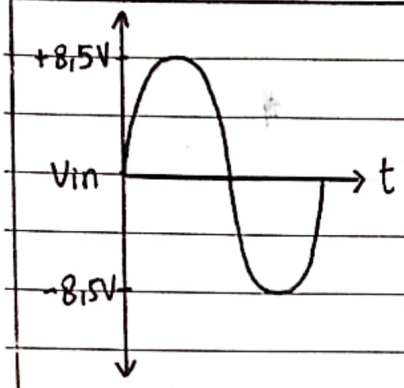
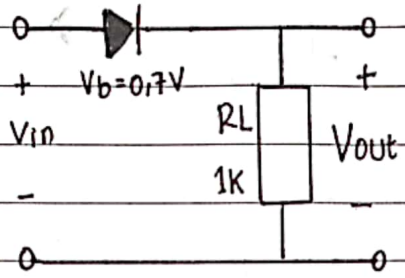
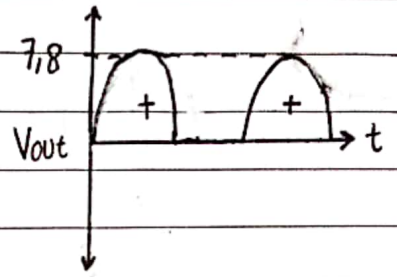
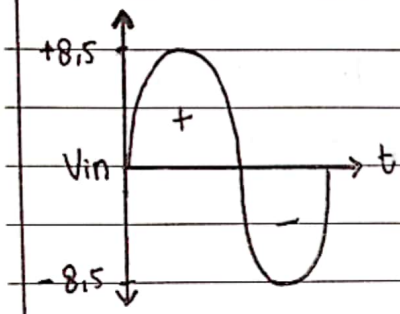
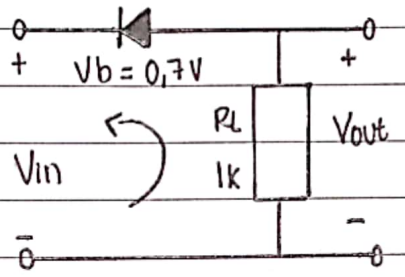
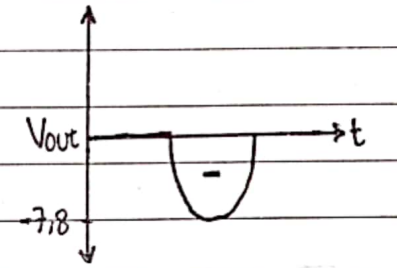
A	B	OUT	Nama Gerbang Logika
0V	0V	0V	
0V	5V	5V	OR
5V	0V	5V	
5V	5V	5V	

A	B	OUT	Nama Gerbang logika
0V	0V	0V	
0V	5V	0V	AND
5V	0V	0V	
5V	5V	5V	

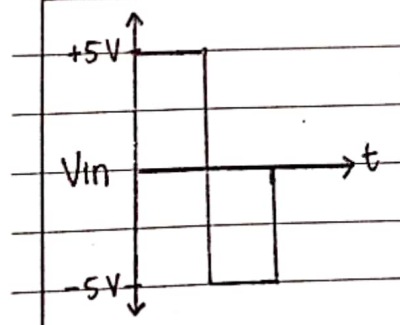
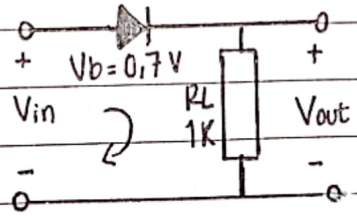
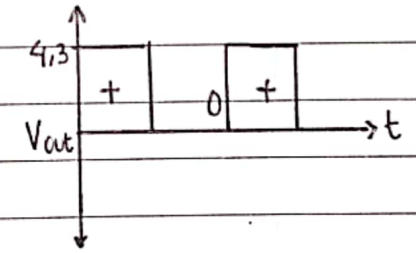
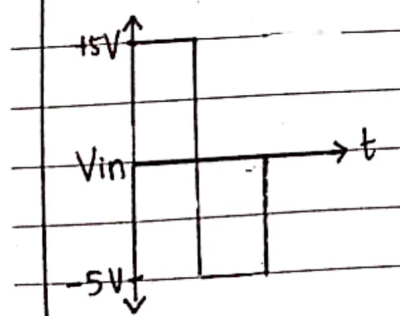
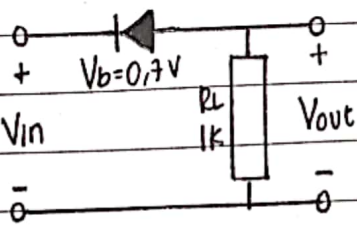
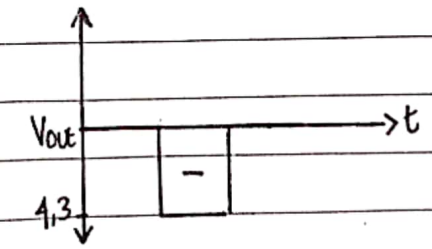
Dipindai dengan CamScanner

No
Date

3. Gambarkan bentuk gelombang keluaran untuk masing-masing rangkaian di bawah ini

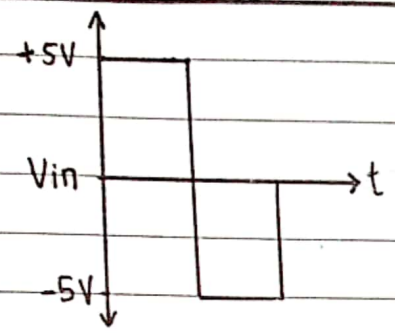
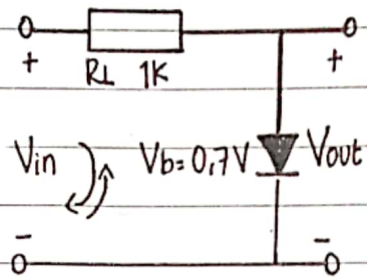
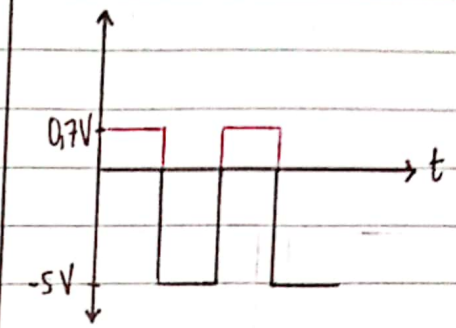
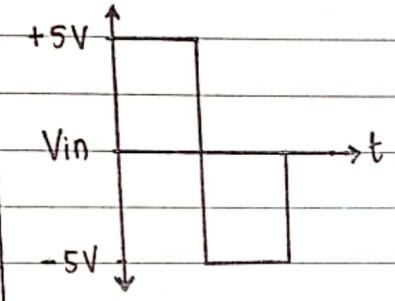
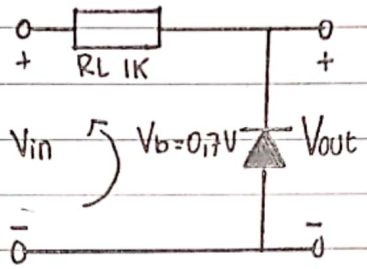
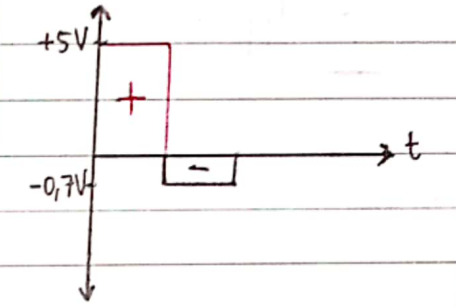
Vinput	Rangkaian Dioda	Vout
	a) 	b) 
		

4. Asumsikan pendekatan dioda dengan mempunyai tegangan ambang Vb = 0,7V. Gambarkan bentuk gelombang keluaran tiap rangkaian bila gelombang masukannya seperti yang terlihat di bawah ini!

Vinput	Rangkaian Dioda	Vout
		
		

Dipindai dengan CamScanner

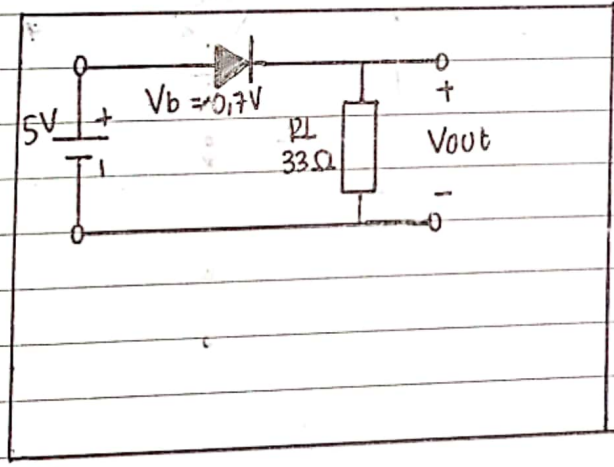
No
Date

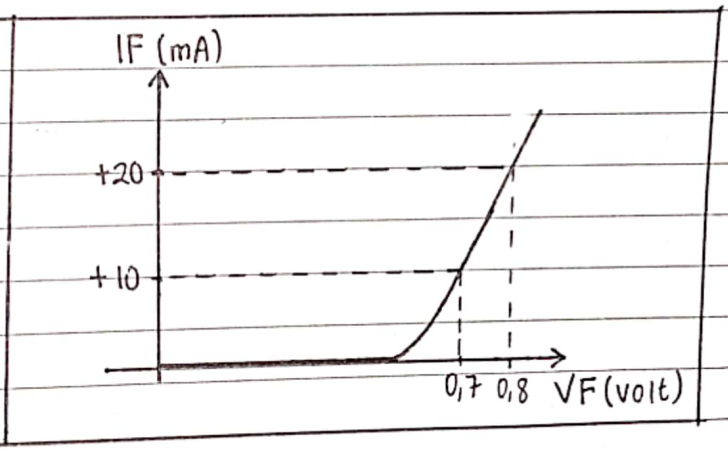
Vinput	Rangkaian Dioda	Vout
		
		

5. Tentukan besarnya IL dan tegangan pada RL bila :

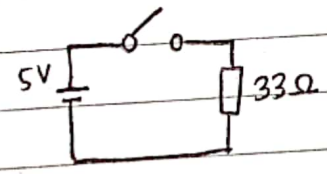
a) Dioda Ideal

b) Dioda tidak ideal dengan kurva karakteristik seperti pada Gambar di bawah ini.



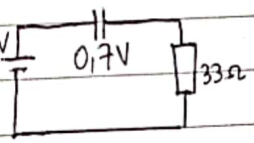


a) Dioda Ideal



$I_L = \frac{5V}{33\Omega} = 0{,}1515A$

$V_L = V_{in} = 5V$

b) 

$V_L = V_{in} - V_b$

$= 5 - 0{,}7 = 4{,}3V$

$I_L = \frac{V_{in} - V_b}{R_L}$

$= \frac{(5-0{,}7)V}{33\Omega}$

$= 0{,}13A$

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